INJIBARA UNIVERSITY

COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF INFORMATION SYSTEM

DORMITORY MANAGEMEN SYSTEM FOR INJIBARA UNIVERSITY

OBJECT ORIENTED PROJECT PROPOSAL

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INJIBARA, ETHIOPIA

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Abstract

This project focuses on the development of a Web-Based Dormitory Management System for Injibara University, aimed at replacing the existing manual dormitory management practices. The current system faces several challenges including data duplication, time-consuming processes, manual errors, and poor accessibility. These issues often result in mismanagement, such as incorrect student allocations, data loss, and increased administrative workload.

The proposed system offers a centralized, automated platform that allows students to register for dormitories, view their room assignments, submit comments or complaints, and access dormitory-related announcements. Administrators and proctors can efficiently manage room allocations, update records, monitor attendance, and generate reports. The system ensures real-time room availability, data accuracy, and role-based access control to maintain security and privacy.

Developed using tools like java and MySQL, the system is both user-friendly and efficient, with a responsive interface for both students and staff. It also supports cost reduction by minimizing paperwork, employee workload, and resource usage. A thorough feasibility study confirmed that the system is technically, operationally, and economically viable.

Overall, the system aims to modernize dormitory management, improve service delivery, and support Injibara University's efforts to adopt digital solutions for enhanced organizational performance.

Background Statement

In the current era, technology is essential for organizations to operate efficiently and meet user needs. However, the dormitory management at **Injibara University** currently relies on a manual, paper-based system that faces several critical challenges.

Student information and dormitory allocations are managed using physical files stored in cabinets, a process that is both tedious and time-consuming due to the large campus population. This manual classification—which considers sex, department, and health status—is prone to frequent human errors, such as the incorrect assignment of male students to female dormitories and vice versa. Furthermore, the lack of a digital database leads to frequent data duplication and leaves vital records vulnerable to total loss from natural or man-made disasters, such as fire.

The operational burden of this manual system is also significant; it requires a high number of employees and a large budget for physical resources like paper, ink, and pens.

CHAPTER ONE

1. INTRODUCTION

Technology is spreading its wing in almost every walks of human life activities. Now a day it is better if every activity is done using new technology in order to fulfill the need of human being, Organization ,Enterprise etc. As today's world there are many organizations and each organizations needs to be preferable, computable and work on fastest way in order to satisfy users interest etc. i.e. they should have facilitate their activities in computerized way .Many developing countries are in a good position to exploit the opportunity of technology revolution and advance human development. The information and communication technology provide new resource materials for expanding communication. Hence, developing the system using technology has a tremendous effect for organizations and offices ;which is in our case the Injibara University Online dormitory management system (INUODMS).Currently, the system is manual based; due to this the students and proctors faces some problems Because of this, we are initiating to develop our project on dormitory system in order to minimize the problem by using computerized system.

1.1. Problem Statement

During the arrangement of students for dormitory allocation, a list of students is received from the student dean, and students are classified based on their sex, level of education, department, and health status. However, during this process, some male students are assigned dorms with female students and vice versa. This issue recurs each year and leads to unnecessary rework. Additionally, all records associated with the overall management process are stored manually on paper and kept in file cabinets. This makes managing and manipulating the data time-consuming and significantly impacts the Dormitory Management System. Manual processing of management activities such as arranging building information for allocation, assigning proctors to buildings, and rearranging students and dorms is tedious due to the large number of students and dormitories on campus and is prone to many problems. All necessary records of these management activities are kept manually on paper and stored in file cabinets. Errors frequently

occur due to manual processing, leading to unnecessary rework. As all dormitory management activities, including assigning proctors, managing dorms, and assigning students to each dorm, are performed manually, the process takes a considerable amount of time. Data duplication often occurs during manual data processing. Data loss can also occur when the storage area is affected by natural or man-made disasters, such as fire. Since everything is done manually by individual workers, the number of employees is high, which in turn requires the university to allocate a large budget for employee salaries. Additionally, a high budget is required for various resources such as copy machine ink, paper, pens, and other materials.

1.2. Objective of the project

1.2.1. General Objective

The general objective of this project is to develop Web Based Dormitory Management System for injibara university.

1.2.2. Specific Objectives

In order to achieve the main objective, we have the following specific objectives:

- To review and analysis the existing systems.
- To gather requirement
- To design a database for the proposed system
- To code the proposed system
- To design a friendly graphical user interface for the proposed system
- To test of the proposed system

1.3. Significance of the System

- The proposed system has the following significances
- Increased accessibility to resources without geographic location or organizational affiliation.
- People can gain access to the information at any time as far as network is available.
- Information resources can be searched easily (like student placement).

➢ Improve the data sharing (uploading and downloading) from one sub-system to another sub-system.

➢ The authorized access to information resources files of the system is more advanced. This means secured login to the system will be developed.

➢ Resources and time saving in system operation and service provision is one of the major benefits.

➢ To manage the students, building and properties information in organized way.

➢ Providing a well-organized and guaranteed record-keeping system with minimum space and effort need.

➢ To enable the university to get acceptance in the outside community.

➢ Developing students effective communication with the university

- Provides quick access to historical data and reports
- Reduces human error and paperwork
- Saves time for both students and administrators

1.5. Scope and Limitations of the System

The scope of the Injibara University Dormitory Management System (INUODMS) is strategically designed to replace manual paper-based processes with an automated, web-based platform. The boundaries of the system are delineated by the specific functional roles and operational constraints outlined below:

Functional Scope (Inclusions)

The system facilitates a centralized environment for managing dormitory lifecycles through distinct actor interactions:

Administrative Governance: The Administrator is empowered with comprehensive management capabilities, including student registration, room and building allocation, and the real-time update of student records. The scope also includes the generation of analytical reports and automated attendance monitoring to ensure data-driven oversight.

Operational Oversight: The Proctor role is primarily restricted to essential supervisory functions. Their access is limited to viewing student lists and accessing dormitory-related data necessary for daily coordination.

Student Information Access: The system provides students with a digital portal for information retrieval. Students can verify their dormitory placement status, view room assignments, and stay informed through institutional announcements.

System Limitations (Exclusions)

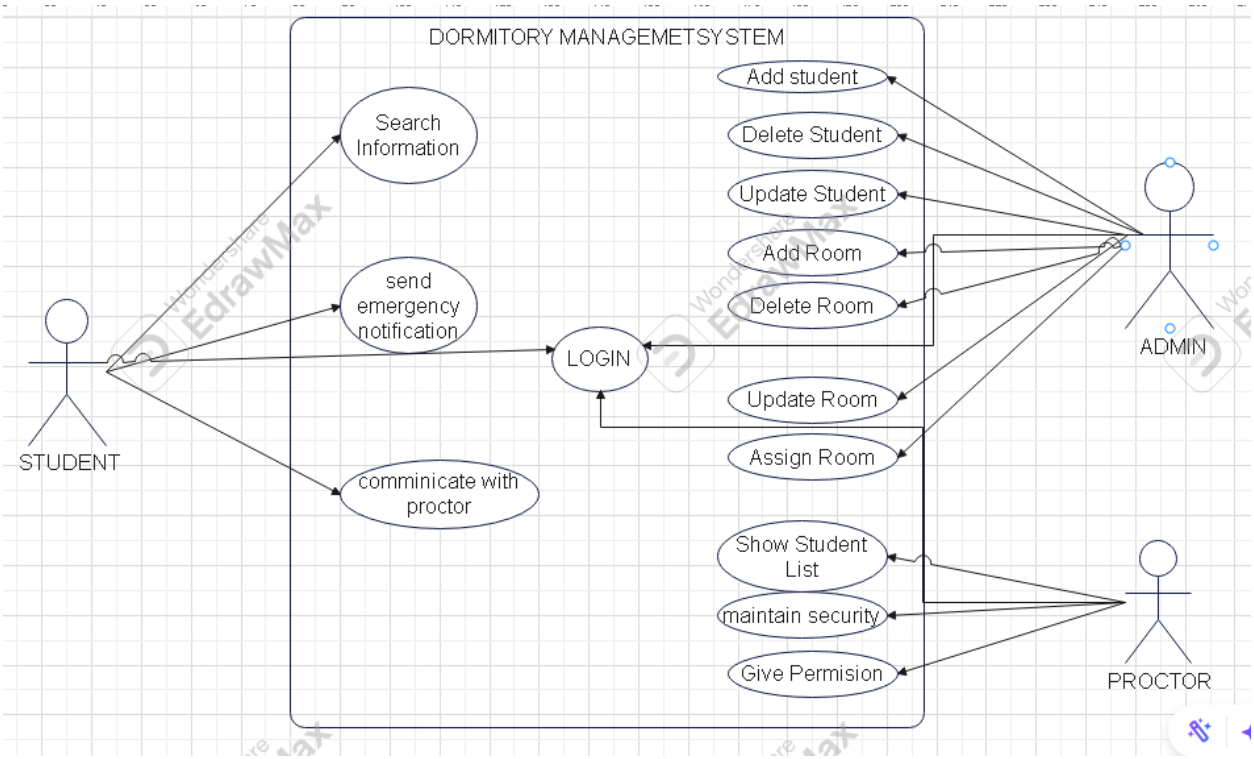
To maintain system focus and security, the following constraints are enforced:

Restricted Communication: Although identified in preliminary diagrams, the current implementation limits the Student actor to information searching only. Students are explicitly unable to send emergency notifications or engage in direct digital communication with the Proctor through this platform.

Authorized Permissions: The Proctor does not have the administrative authority to grant formal permissions (such as leave requests) or manage advanced security protocols within the system.

Connectivity and Infrastructure: As a web-based solution developed using Java and MySQL, the system requires active network connectivity and does not currently include integration with external hardware for physical security or biometric entry.

1.4. USE CASE DIAGRAM



Conclusion

The development of the Web-Based Dormitory Management System (INUODMS) marks a significant transition for Injibara University from an inefficient manual process to a modern, automated solution. By addressing critical issues such as data duplication, human errors in room assignment, and the physical vulnerability of paper records, the system ensures a more accurate and secure management environment.

The integration of Java and MySQL provides a robust platform that enhances operational efficiency, reduces administrative costs related to physical resources, and streamlines communication between students, proctors, and administrators. Ultimately, this project successfully fulfills its objective of providing a technically and economically viable system that supports the university's digital transformation and improves overall service delivery for the student population

THANK YOU!